

**IS - NPMD -
DERMATOLOGY**

Soft Tissue Infections- Wounds And Burns

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LEARNING & TESTABLE OBJECTIVES

1. List the major pathogens of burns infections and wound infections

For each of the pathogens listed in the bug table,

2. Describe the morphology, cultural and identification characteristics

3. Describe the relevant epidemiology including reservoir/ source and mode of transmission

4. Associate it with disease presentation and possible complications

5. Correlate the virulence factors with the pathogenesis of the disease

6. Describe the laboratory diagnosis

7. Select the antimicrobial of choice and describe its mode of action

8. Choose the appropriate preventive/ prophylactic measures

Pathogen	Disease
<i>Pseudomonas aeruginosa</i> <i>Staph aureus</i> and <i>S. epidermidis</i>	Burns infections
<i>Clostridium perfringens</i> and other clostridia	Gas gangrene/ Myonecrosis
<i>Streptococcus pyogenes</i>	Necrotizing fasciitis
Non sporing anaerobes: <i>Bacteroides</i> , <i>Prevotella</i> , <i>Fusobacterium</i> , <i>Peptococcus</i> , <i>Peptostreptococcus</i>	Deep seated infections
<i>Vibrio vulnificus</i>	Cellulitis and abscess
<i>Erysipelothrix rhusiopathiae</i>	Seal finger/ Whale finger/ Erysipeloid
<i>Eikenella</i>	Human–bite associated infection
<i>Capnocytophaga</i>	Dog-bite associated infection
<i>Pasteurella multocida</i>	Cat bite/ dog bite associated infection
<i>Bartonella henselae</i>	Cat Scratch disease

MAJOR PATHOGENS

Traumatic Wounds

- *P. aeruginosa*
- *Clostridium* spp
- *S. pyogenes*
- Sea water – *Vibrio vulnificus*
- Seal finger-*Erysipelothrix*

- **Bites** – *Pasteurella*, *Eikenella*, *Capnocytophaga*
- **Cat scratches** – *Bartonella*

Burns

- *P. aeruginosa*
- *S. aureus*
- Enterobacteriaceae

Opportunistic Infections

- Anaerobes-non sporing
- *S. aureus*
- *P. aeruginosa*
- Enterobacteriaceae

Pseudomonas aeruginosa

SKIN INFECTIONS

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Burns infection

The most common cause of burns infections is *Pseudomonas*.



Wound infections especially Diabetic foot infections. Colonization of shoes/ contamination of wound from environmental source

Pseudomonas aeruginosa

SKIN INFECTIONS



Hot tub Folliculitis

Community acquired infection following exposure to contained contaminated water (**whirlpools, swimming pools, water slides, bathtubs**), use of diving suits etc.



Ecthyma gangrenosum

- Infection of the skin often seen in immuno-compromised patients such as those with neutropenia
- **Lesion: round/ oval with a halo of erythema and necrotic center**
- Single or multiple and heal with scar formation.
- Invasion of blood vessels by *Ps. aeruginosa* causing infarction and necrosis. (**Exotoxin A**)

Predisposing conditions:
IMMUNOSUPPRESSED host
CHEMOTHERAPY
NEUTROPENIC

Pseudomonas aeruginosa

- Gram negative bacilli, aerobic
- Oxidase positive
- Fruity odor (grape –like)
- Pigment production-pyocyanin (fluorescein), pyoverdinin etc.

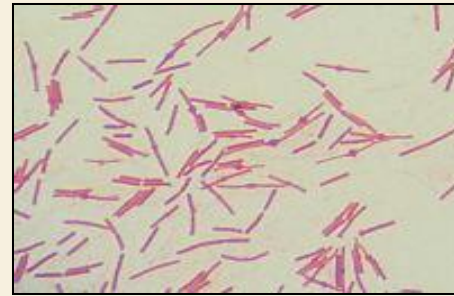
Virulence factors

Enzymes: Elastases, Phospholipase C.

Toxin: Exotoxin A – acts by inhibiting protein synthesis by ADP ribosylation of EF-2.

Treatment

- Piperacillin + aminoglycoside
- Multidrug resistant strains emerging.
- **Prevention**- avoid flowers in hospitals. Avoid moisture and environmental source contamination.



Gram
Negative
Bacilli



Nutrient agar
showing green
pigmented
colonies

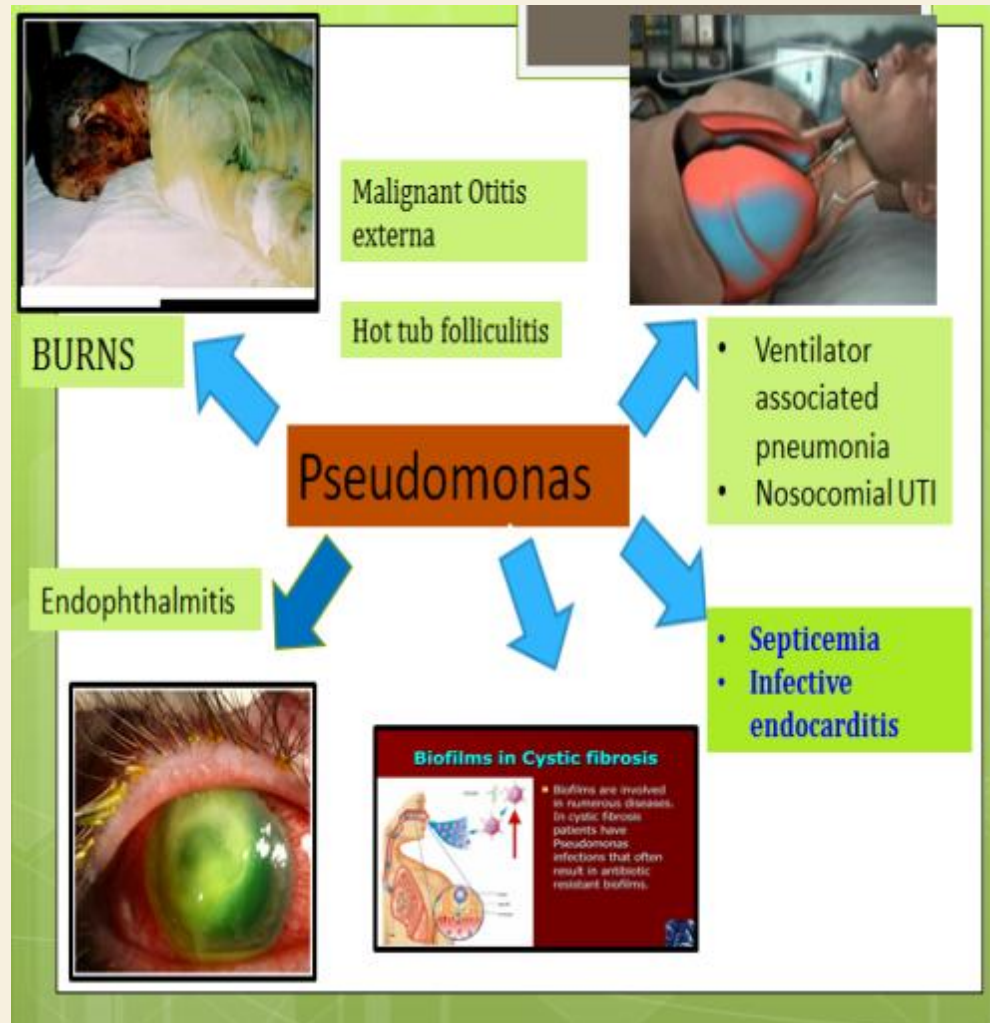


Mac Conkey's agar
showing non lactose
fermenting (NLF) pale
colonies



PSEUDOMONAS AERUGINOSA INFECTIONS (THE BIG PICTURE)

- Opportunistic pathogen causing:
 - **P** = pneumonia, especially in cystic fibrosis patients and patients on ventilators
 - **S** = sepsis
 - **E** = external otitis (swimmer's ear)
 - **U** = urinary tract infections
 - **D** = drug use and diabetics
 - **O** = osteomyelitis
 - Burn and wound infections
 - Nosocomial infections
 - Hot tub folliculitis





OBLIGATE ANAEROBES

CLOSTRIDIA

BACTEROIDES

FUSOBACTERIA

PEPTOCOCCUS

PEPTOSTREPTOCOCCUS

Gas Gangrene

Bacterial/ Clostridial Myonecrosis

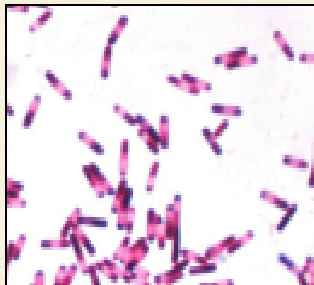
- Clostridial myonecrosis, or **gas gangrene**, is a fast-spreading and potentially life-threatening necrotic **infection caused due to the multiplication of bacteria (mainly Clostridium spp).**
- **Causative agent- multifactorial, in which *C. perfringens* plays the predominant role.**

- **Gram positive bacilli**, with sub-terminal spores that are resistant to heat, drying and several chemicals. (Sensitive to Glutaraldehyde, formaldehyde and Ethylene oxide)

- Non-motile, obligate anaerobic

- Form several toxins and enzymes.

- 12 different toxins formed out of which an important toxin is **Alpha toxin/ Phospholipase C (lecithinase)- detected by Nagler's reaction/ Immunoassays**



Gram stain of culture smear of *C. perfringens* showing subterminal spores

- Anaerobic infections are associated with foul smelling pus.
- They lack enzymes- Superoxide dismutase and catalase hence die in the presence of O₂

PATHOGENESIS AND CLINICAL FEATURES OF GAS GANGRENE

Predisposing factors

- Road Traffic injuries, Puncture wounds, Gunshot wounds, War injuries, Compound fractures.

Rarely:

- Following surgeries like amputation for peripheral vascular diseases .
- Following injection of Adrenaline.



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Type A *Cl. perfringens* inhabits intestinal tracts of humans and animals and is widely distributed in nature, esp. in soil and water contaminated with feces.

Introduction of Clostridial spores by trauma or surgery

Low redox potential in damaged tissues creates anaerobic environment, spores germinate and vegetative forms multiply

Produce toxins: **Lecithinase (α -toxin)**, **θ toxin**, **collagenases**, **hyaluronidases etc.**

Extensive damage to tissues, cell membranes, RBCs, leukocytes, platelets with myonecrosis, cellulitis, increase in vascular permeability and extravasation.

- Extensive damage to tissues cell membranes and muscles.
- Abundant gas formation/ crepitus can be felt. This further reduces the blood supply, extending the area of anoxic damage.
- Increase in **vascular permeability** and extravasation can lead to shock, renal failure and death.

Laboratory diagnosis

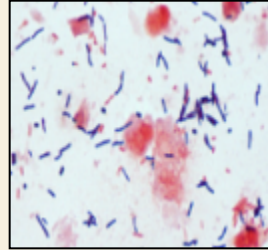
Treatment/ Prevention

Sample: Tissue biopsy/ deep wound swabs.

Microscopy- Grams stain of wound swab- Scanty inflammatory cells and GPB.

Culture: Blood agar (anaerobically incubated): Double zone hemolysis.

- Stormy fermentation in milk media:-acid and gas formation in Litmus milk medium
- Confirmatory tests: Nagler's reaction on egg yolk agar



Neutralization test for the detection of activity of α -toxin also called Lecithinase or Phospholipase C.

Demonstrated on Egg Yolk Agar Lecithinase (α -toxin; phospholipase) released by the clostridial strain, hydrolyzes phospholipids in egg-yolk agar around streak on right (seen as a pearly opacity).

Antibody against α -toxin inhibits activity around left streak due to neutralization of the toxin.

Treatment

- Aggressive treatment with surgical debridement /amputation if needed.
- High -dose Penicillin therapy OR Penicillin+ Clindamycin combination.
- Hyperbaric oxygen therapy

Prevention

- Good wound care
- Use of prophylactic antibiotics

NON-SPORE-FORMING ANAEROBES

GPR

Actinomyces spp.
Propionibacterium acnes



GNR

Bacteroides spp.
Fusobacterium spp.
Prevotella melaninogenica

GPC

Peptococcus
Peptostreptococcus

NON-SPORING ANAEROBIC ORGANISMS

Part of **normal flora**; **Opportunistic** pathogens

Infection occurs when host defenses and anatomical barriers are altered (tissue ischemia, trauma, surgery, perforation of intestine...)

Often mixed infections: the growth of facultative anaerobes facilitates obligate anaerobic growth

- ***Bacteroides fragilis***

- **Gram negative bacilli**
- **Predominant gut anaerobe**
- **Intra-abdominal** and soft tissue infections **below the waist**
 - Surgical wounds
 - Decubitus ulcers
- Septicemia
- Antibiotic resistance is common

Anaerobic Gram-Positive Cocci ***Peptococcus* and *Peptostreptococcus***

Brain abscesses, sinusitis,
intraabdominal infections, skin and
soft tissue infections
Aspiration pneumonia

- ***Prevotella melaninogenica* & *Fusobacterium***

- **Gram negative bacilli**
- *Prevotella* – GI & oral flora
- *Fusobacterium* – GI, oral, upper RT flora
- **Infections above the waist**
 - Sinuses, ear and periodontal infections
 - Brain abscesses
 - **Aspiration pneumonia (alcoholics)**

Diagnosis- Growth under anaerobic conditions, biochemical testing and gas chromatography

Treatment

Metronidazole

Most produce beta-lactamases
Surgical drainage

Necrotizing fasciitis

Most common agent- *Strep pyogenes*

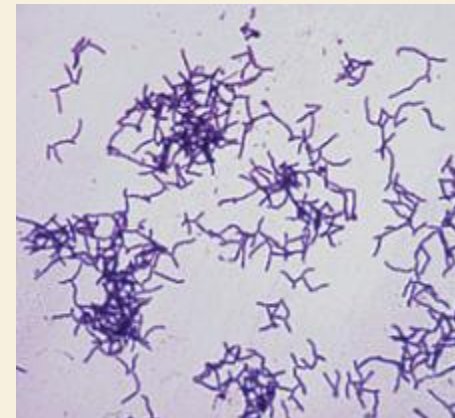
- A severe, destructive lesion involving the fascia and destruction of underlying muscle and fat. Caused by **flesh eating bacteria**-may be mixed/**usually due to Group A Strep (GAS).**
- Can cause death due to multiple organ failure.
- **Rapid spreading growth of streptococci and action of SPE toxin.**
- Differential diagnosis- Gas Gangrene



Erysipelothrix rhusiopathiae

Erysipeloid/ Whale finger/ Seal finger

- Violaceous lesions caused by *Erysipelothrix rhusiopathiae*
- Wounds caused due to fishing / handling whales/ seals/ pigs.(Occupational disease)
- **Gram positive bacilli** that are catalase negative.
- Treatment: Penicillin



Vibrio vulnificus

Found in **salt water**; may contaminate **oysters**



Cuts when opening oysters

Wound infection ➡ invasion ➡ necrotic cellulitis



- **Virulence factors**
 - Capsule, cytotoxic enzymes, LPS
- **Diagnosis**
 - Microscopy – **Gram negative curved bacilli**
 - Culture
 - **Halophilic**: requires salt for growth
 - **Thiosulfate Citrate Bile Salts Sucrose agar (TCBS)** – selective and differential alkaline media- usually greenish colonies.
 - Identification
 - Oxidase positive, ferments lactose (unlike other vibrios)
- **Treatment** – Doxycycline + 3rd Gen Cephalosporins

WOUND INFECTIONS

ANIMAL BITES/ SCRATCHES

- **Bites** – *Pasteurella, Eikenella, Capnocytophaga*
- **Cat scratches** – *Bartonella*



Bartonella henselae

causes

CAT SCRATCH DISEASE

Gram -negative rods which do not stain well with gram's stain

Better stained with silver stains such as Warthin-Starry stains.

Usually do not grow in culture

BARTONELLA- CLINICAL PRESENTATION/ DIAGNOSIS/ TREATMENT

Immuno-competent host:

- ✓ Benign disease manifesting as **local papule and regional lymphadenopathy**.
- ✓ Entry of organism into regional lymph nodes via the bloodstream causes **chronic regional lymphadenopathy**.
- ✓ **A granulomatous host reaction.**
- ✓ **Bacteria may be seen in lymph nodes but are very scanty.**



Hand showing local papules at the site of bite



Severe axillary lymphadenopathy

Immuno-compromised host:

- ✓ Severe spectrum of diseases.
- ✓ Bite results in a **local papule** followed by **extensive multiplication of Bartonella with bacteremia and involvement of lymph nodes, skin, liver and spleen.**
- ✓ **Fevers with bacillary angiomatosis** (vascular proliferative disease of the skin) manifesting as blood filled nodules and **bacillary peliosis** (hepatitis).



Skin lesion of bacillary angiomatosis in AIDS patient with *B.henselae* infection

Diagnosis:

- ✓ **Silver stain/Serological tests/PCR**

- ✓ **Cultures are usually sterile**

Treatment: **Azithromycin / Doxycycline**



Pasteurella multocida - associated with bites of cats/ dogs.

Gram negative rods/
cocco-bacilli.

Non-motile

Facultative anaerobe

Grow on Chocolate and
Blood Agar

No growth/ poor growth
on MacConkey's agar

Oxidase +

Virulence factors:
Capsule; Endotoxin

Source of infection-Oral cavity / upper respiratory
tract of cats and dogs.

Transmission- scratch/
bite/ salivary contamination

Wound infection

Cellulitis

Lymphadenitis

Pulmonary dysfunction in immuno-compromised.

Sometimes, Septic arthritis/Osteomyelitis

✓Treatment

✓Thorough cleaning of wounds

✓Amoxycillin + Clavulanic acid

✓Macrolides are usually not given(due to
resistance)

OTHER BITE- ASSOCIATED INFECTIONS

HUMAN BITE

- *Eikenella corrodens*

Part of HACEK group —
cardiovascular infections

- Corrodes agar
- Bleach-like odor
- Human bites or fist fight injuries
- Cellulitis



DOG BITE

- *Capnocytophaga canimorsus*

- GNR
- Dog bites
- Cellulitis



Thank
you!!